Survival Analysis

Seminar Project

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Sessions

Monday 16th

Survival Analysis Fundamentals

core concepts
(2 hour sesion)

Wednesday 18th

Applications and Projects

exploring real-world uses
(1 hour sesion)

Monday 23rd

Coding a Real App

hands-on coding session
(2 hour sesion)

Wednesday 25th

Seminar Project Organization

project planning
(1 hour sesion)

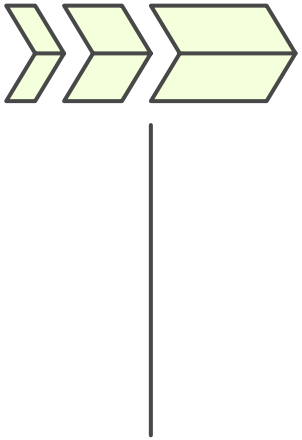
TBD

Seminar Project Presentations

Final presentations of student projects

Sessions

Wednesday
25th



Seminar
Project
Organization

project planning
(1 hour sesion)

Index

Objective

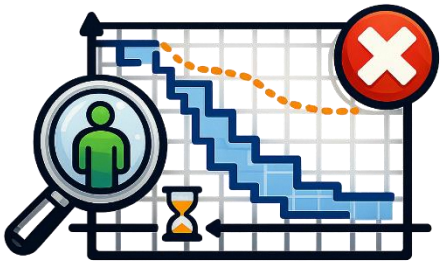
Organization

Topic options

Any questions?

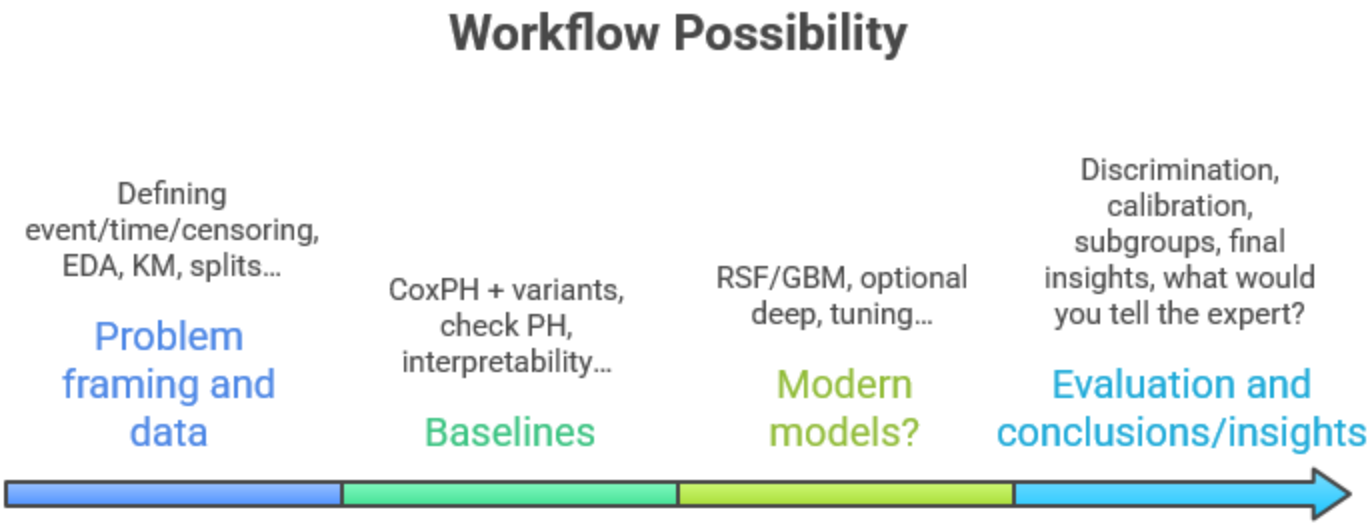
Project Objective

Goal: Survival Analysis in a real Use Case



- **Implement Use Case**
- **Compare Models**
- **Draw Conclusions**

Project Organization

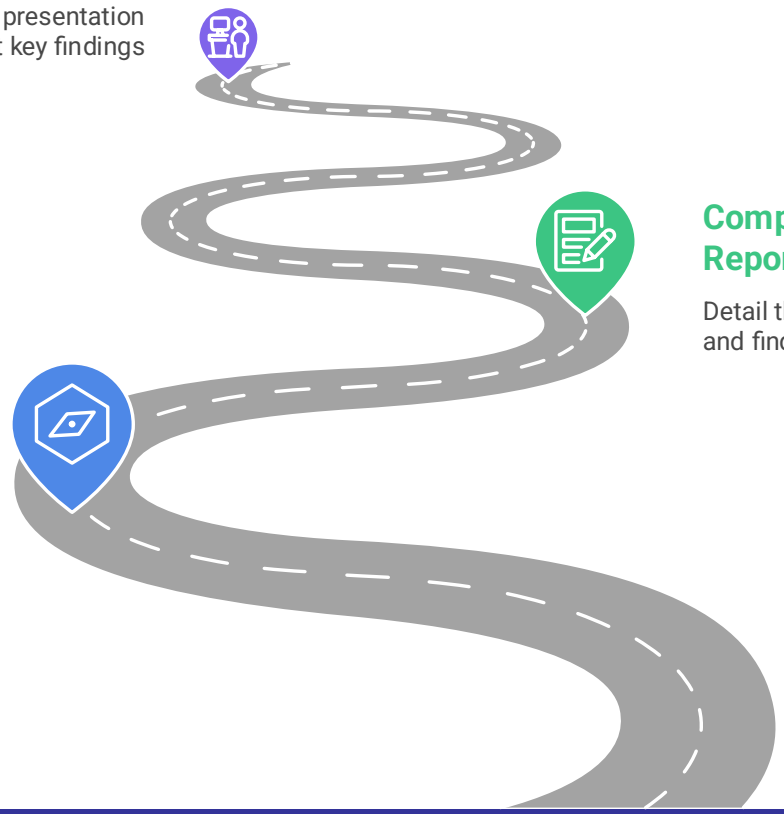


Project Deliverables

Presentation date to be decided…

Presentation Delivery

5-minute presentation about key findings



Comprehensive Report Writing

Detail the project process and findings

Git Repository/Notebook Creation

Establish a repository or notebook for project code

Tips

- Use existing libraries (**lifelines**, **scikit-survival**, **pycox**)
- Survival-specific preprocessing list:
  - ✓ Define T and E/δ clearly
  - ✓ Handle categorical variables consistently
  - ✓ Missing values: imputation or missingness indicators
  - ✓ Splits
- For different models, use Optuna or GridSearch for tuning (keep it small!)
- Add a baseline check: KM curve + predicted curves for “high vs low risk” profiles
- For interpretability:
  - ✓ Cox: HR table + confidence intervals
  - ✓ Tree-based model: feature importance
  - ✓ Deep: risk stratification plots
- Don’t report only C-index!

Survival Datasets

- **Telco Customer Churn:**
  - About Dataset

- **Breast Cancer:**
  - About Dataset

Survival Datasets

- **SurvSet: An open-source time-to-event dataset respository**
  - Repository
  - 77 datasets
  - References: *Drysdale, E. (2022). SurvSet: An open-source time-to-event dataset repository. arXiv preprint arXiv:2203.03094.*

- **Lifelines**
  - Repository
  - 27 datasets
  - References: *Davidson-Pilon, C. (2019). lifelines: survival analysis in Python. Journal of Open Source Software, 4(40), 1317.*

Survival Datasets

- **Your turn:** Feel free to explore datasets that match your interests!

- Where to search?
  - Kaggle, GITHUB...
  - **Some ideas:**
    - **Game of Thrones:** Predict the "Time-to-Death" of characters based on their house, battles... About Dataset
    - **NASA Engineering:** Estimate the Remaining Useful Life (RUL) of turbofan engines to prevent catastrophic failures. About Dataset